

BUILD WITH AI HACKATHON

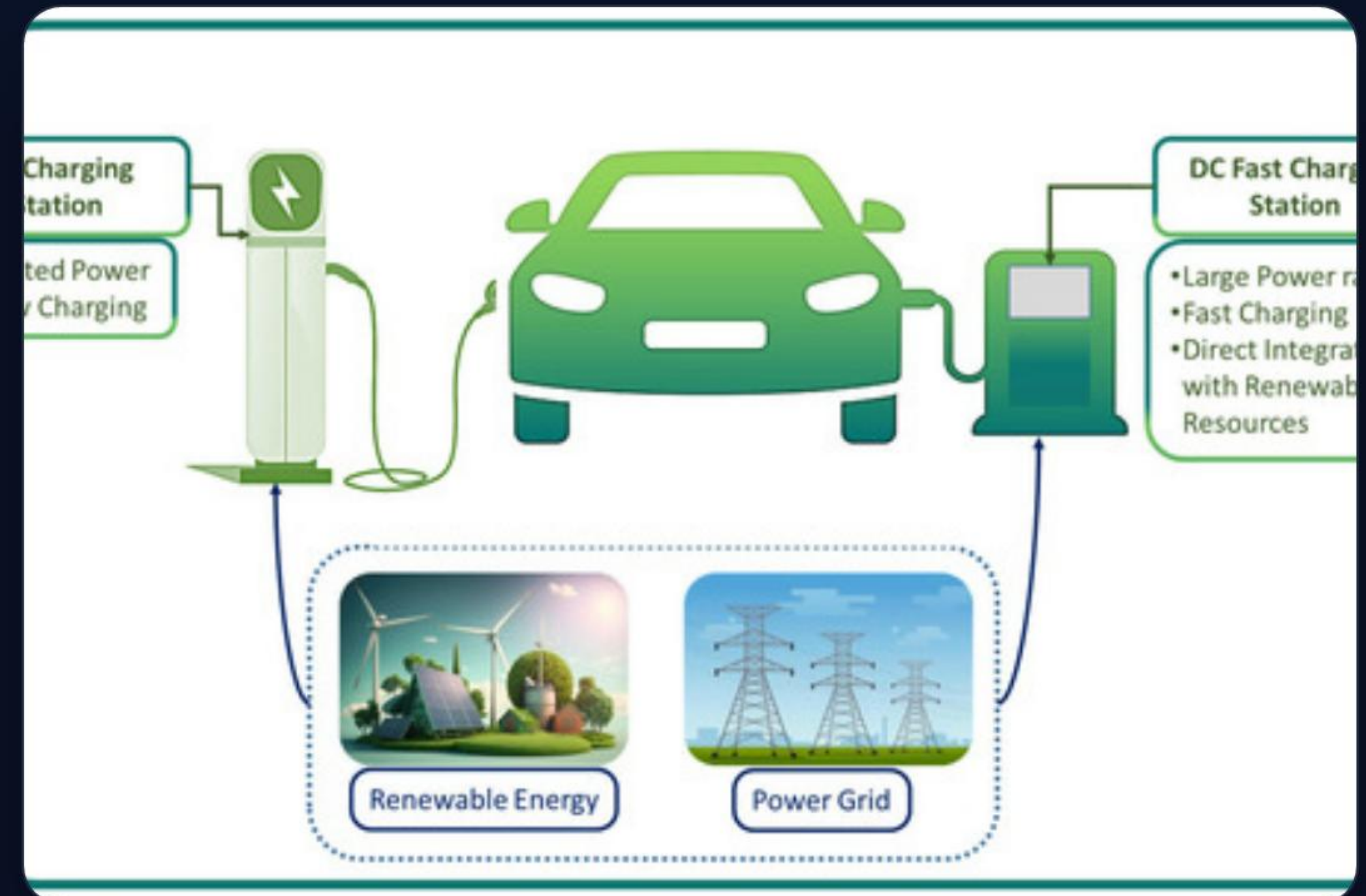
EcoDrive AI

AI-Driven Engine Compatibility Diagnostics, Performance Analytics, and Preventive Optimization for India's E20 Bio-Fuel Transition.

| 🎯 Problem Statement

India's mandate targeting 20% ethanol blended petrol by 2026 creates silent mechanical risks for 300 Million+ legacy vehicles:

- **Mechanical Corrosive Oxidation:** Phase-separated water triggers rapid internal galvanic rust in steel fuel tanks.
- **Elastomer Material Failure:** Standard fuel lines in pre-2023 cars and bikes break down under intense aromatic stress.
- **Anomalous Lean Burn Conditions:** Lower calorific fuel increases Exhaust Gas Temperatures (EGT), accelerating engine failure.
- **The Diagnostic Gap:** Generic lookup tables cannot predict real-time component degradation or provide localized remedial actions.



| Our Solution



1. OBD-II Telemetry

Monitors vital engine indicators, tracking changes in Short-Term Fuel Trim (STFT), Exhaust Temperatures, and O2 sensors to identify lean ethanol fuel shifts before damage occurs.



2. ML Diagnostic Engine

Processes real-time sensor parameters through a Random Forest Classifier to identify patterns of degradation and predict corrosion or misfire risks before hardware failure occurs.



3. Actionable Remediations

Recommends specific, localized actions (such as upsizing carburetor jets or upgrading fuel hoses to Viton synthetics) to prevent high repair costs.

| Technology Stack & Partner Integrations

Production AI Stack

Built on robust data-science frameworks matching our Jupyter execution architecture:

- **Predictive Analytics Core:** Optimized Random Forest Classifiers built with Scikit-Learn to capture non-linear engine anomalies.
- **Feature Engineering:** Custom metrics tracking local thermal indices (EGT / Engine_CC) to isolate small-engine motorcycle wear.
- **Data Pipelines:** Clean Pandas and Numpy structures configured for direct micro-controller deployment.

Hackathon Partner Integrations

Maximizing score metrics using advanced technology partner APIs:

- **Google Gemini API (LLM Integration):** Translates raw diagnostic code alerts into clear, multi-lingual voice guides and practical advice for local mechanics.
- **Google Cloud Run:** Scales the lightweight, serverless telemetry-parsing engine to process thousands of OBD-II signals concurrently.
- **Android/Flutter Edge App:** Pulls real-time engine metrics from vehicle sensors via Bluetooth OBD-II receivers.

Market Opportunity & Value

75%

Maintenance Savings

High-Impact Solution for India's Clean Energy Shift

EcoDrive AI delivers measurable economic support directly aligned with the core criteria of the Build-with-AI Hackathon:

- **Target User Base:** Over 250 Million legacy 2-wheelers and 50 Million older family cars in India seeking practical guidance on fuel compatibility.
- **Socio-Economic Impact (40% Weight):** Reduces unexpected engine failure and repair costs for middle-class vehicle owners, making the transition to cleaner fuels easier.
- **Operational Working MVP (30% Weight):** Built with an end-to-end Python engine, ready to run directly in lightweight edge-computing environments.